TEXORA · LECTURE SHEET · COURSE GUIDE

Industrial Visit: Report Writing and Presentation Guide

How to turn one factory visit into the full 100 marks of TE-3110: what to observe in knitting, dyeing and the testing laboratory, the data to collect on the floor, and the report and presentation the marks are paid for.

TE-3110 · INDUSTRIAL VISIT · GUIDE SHEET · LEVEL 3 TERM 1

1. The course in brief

TE-3110 (credit 1.0) requires every student to **visit at least one textile industry** for practical experience of a working mill. There are no lectures — the factory is the classroom, and the evidence you bring back is the coursework. This second visit carries a higher expectation than Level 2: you have now studied knitting, textile testing and quality evaluation, and coloration theory, so the examiner expects you to *measure, calculate and evaluate*, not merely describe. The marks distribution is fixed by the syllabus:

COMPONENT	WEIGHT	WHAT EARNS THE MARKS
Visit report	**60 %**	A complete, well-structured report based on your own observations, data and calculations (Section 5)
Presentation	**40 %**	A clear, timed talk on the visit, defended in viva questioning (Sections 6–7)

Report 60%, Presentation 40% — the visit must be *worked*, not just attended, with a notebook, question list and calculator (Sections 3–4).

2. Before and during the visit — working rules

- **Prepare the question list in advance** from Sections 3–4, customised to the mill type. Review the theory the night before — knitting and loop formation, test methods and standards, dye classes, pretreatment and finishing — so questions go deeper than naming machines. Carry a notebook, pen and calculator; ask before photographing, as many mills prohibit it.

- **Safety and dress:** follow the mill's rules exactly — no loose clothing or open hair near machinery, needles and stenters, stay with the guide, do not touch running machines, settings, chemicals or materials; keep clear of dosing lines and hot surfaces in the dye house.
- **Record numbers, not impressions.** "A large knitting section" earns nothing; "46 circular machines, 30 in. diameter, 24 gauge, 38 working, efficiency about 82%" is report material. Mark verbal figures as reported, distinct from what you counted yourself.
- **Note the flow, then test it.** In every department ask: what comes in, what is done, what goes out — and at Level 3 add: *how is its quality checked, and against what limit?* Collect one worked example per major department: one knitting machine in full, one dyeing lot, one laboratory test with result and standard.

3. What to observe, department by department

Observe the departments the mill actually has. For each, record machines (Section 4), process sequence, production figures, and its quality checks.

DEPARTMENT	WHAT TO OBSERVE
Yarn & chemical store	Yarn types, counts and lots, traceability, receipt testing and issue; dye classes held, chemical storage and labelling
Knitting / weaving (if present)	Knitting: circular, flat-bed and warp machines — diameter, gauge, feeders, rpm, fabric in production (jersey, rib, interlock, fleece), stop causes (yarn break, needle break, hole). Weaving: preparation (warping, sizing), then looms — type, construction, speed, efficiency
Dyeing and finishing (if present)	Follow one lot: pretreatment (desizing, scouring, bleaching, mercerising), dyeing machine (jet, soft-flow, jigger, pad) with liquor ratio, temperature profile and dye class, washing-off, finishing (stenter, sanforising, calendaring); shade approval procedure and reprocessing rate
Garment (if present)	Marker and cutting, sewing lines (layout, output, efficiency), finishing, pressing, inspection, audit and packing
Quality control laboratory	Give this the longest stay. Fibre tests (length, fineness, strength); yarn tests (count, CSP, U%, imperfections, hairiness); fabric tests (GSM, tensile, tear, bursting, pilling, dimensional stability, colour fastness, spectrophotometer). For each: instrument, standard (ISO, ASTM, AATCC or buyer) and one result or limit
Compliance, ETP, utilities, dispatch	Order planning; fire/building safety, PPE, welfare, audits; ETP type (physico-chemical, biological or combined), capacity, inlet/outlet testing, sludge route; boiler, generator, captive power; finished-goods inspection, packing and dispatch

4. Data to collect — the checklist

DATA GROUP	ITEMS TO CAPTURE
Company facts	Name, location, year established, mill type, workforce, main products, buyers if disclosed
Machines, materials, counts	Machine name, maker/country, number installed and running, key specification — knitting: diameter, gauge, feeders, rpm; dyeing: type, capacity (kg), liquor ratio; stenter: width, temperature, speed. Yarn counts, blends, constructions (courses/wales, loop length, GSM), dye classes
Production and efficiency	Daily/monthly production as reported, shifts per day, efficiencies, output for your worked example, dyeing right-first-time and reprocessing rates if disclosed
Quality & environment	Checks per department, one lab result set with standard and limit, common faults (knits: holes, needle lines, barre; dyeing: uneven shade, fastness failure), fate of rejects; ETP data as in Section 3

Attempt two checks from your own figures before writing. Machine efficiency compares actual with theoretical output:

$$\textit{Efficiency (\%)} = (\textit{Actual production} \div \textit{Theoretical production}) \times 100$$

Also check a reported GSM against loop length, stitch density and count. If a figure is withheld, record that — "right-first-time rate not disclosed" is an accurate observation; inventing a number is the one unforgivable error in a visit report.

5. The report — structure

The report follows the material through the mill. Build the process-flow diagram first (Figure 1); every chapter then expands one box of it.

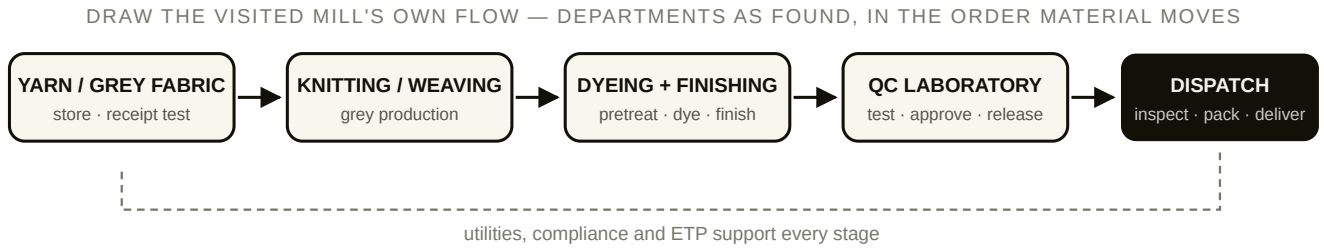


Figure 1 — Generic process flow of a textile mill. Replace each box with the visited mill's actual departments and machine lines, in the order material moves; omit departments the mill does not have. Show the laboratory as a control loop over production.

1. **Title page** — title, course code TE-3110, your name, roll/ID, institution, industry visited, dates of visit and submission.
2. **Acknowledgement** — brief thanks to the institute, course teacher, and the mill's management and guide by name and role.
3. **Table of contents** — chapters with page numbers.
4. **Introduction** — purpose of the visit, stated as questions (e.g. how shade is controlled, how quality is proved).
5. **Company profile** — the company facts of Section 4: identity, type, products, capacity, organisation.
6. **Process flow** — Figure 1 adapted, with a paragraph on the material's route and laboratory control points.
7. **Department-wise observations** — the core: one section per department (Section 3), with data from Section 4 and one worked example per major department.
8. **Data and calculations** — efficiency calculation, GSM or production check, and laboratory results with standards and limits, each commented. Show the working; a result without working earns little at Level 3.
9. **Compliance and environment** — ETP and its testing, safety and welfare provisions, and the buyer pressure behind them.
10. **Findings** — explicit theory-to-floor links (e.g. "the effect of loop length on GSM was seen when…"), mill strengths, anything unexpected.
11. **Limitations and conclusion** — departments not shown, data not disclosed, short duration; then what you can now understand that you could not before.
12. **References** — textbooks, course sheets, test standards cited, and company literature provided.

> **REPORT DISCIPLINE**
>
> Write the department sections within two days of the visit. Every number must trace to a notebook entry — observed, read from a display, calculated by you (show the working), or marked "as reported by the guide". Include photographs only if the mill permitted them.

6. The presentation — structure and slide plan

The presentation compresses the report into a timed talk (confirm the minutes with the course teacher; plan for about 8–10 minutes). One idea per slide; slides carry diagrams, tables and permitted photographs — the speaking carries the explanation.

SLIDE	CONTENT	SPEAKING POINT

1–2	Title and company profile — course, industry, your name/ID; type, products, scale	Introduce the visit; place the mill in the industry
3	Process flow diagram	Walk the audience along the material's route, naming laboratory control points
4–5	Knitting/weaving and dyeing deep dives — one worked example each	Specification, inputs and outputs, one number you checked
6	QC laboratory — tests, standards, one result set	How quality is proved: test, standard, limit, failure consequence
7	Production and efficiency — your calculation slide	Your figures with summarised working, and what they show
8	Compliance and ETP	What the plant does, what is tested at outlet, why buyers require it
9–10	Findings, limitations, conclusion, questions	Theory seen in practice; what you could not see; invite questions

- **Rehearse aloud, timed.** Most marks are lost by talks cut off mid-way; a rehearsed 9 minutes beats an unrehearsed 15. **Know your own numbers** — every figure on a slide may be questioned in the viva; if you cannot explain its source and working, remove it. **Prepare from Section 7.**

Glossary

Right-first-time (RFT) — share of dyeing lots meeting shade and quality without correction. **Loop length** — yarn length in one loop; with stitch density and count it controls GSM. **Count-strength product (CSP)** — yarn count × lea strength. **Unevenness (U%)** — percentage mass variation of a yarn; high U% predicts barre and breaks. **BOD / COD** — oxygen-demand measures of effluent pollution. **Compliance audit** — buyer inspection against safety, labour and environmental codes, on which orders depend.

Key points

• TE-3110 (credit 1.0) marks: Report 60 %, Presentation 40 % — at Level 3 the visit must be measured and evaluated, not merely described.
• In every department ask: what comes in, what is done, what goes out, and how its quality is checked and against what limit.

• Give knitting, dyeing and the QC laboratory the deepest attention — full specifications, one dyeing lot in detail, laboratory results with standards. Record numbers with sources, show calculations, and build report and presentation around the same flow diagram.

7. Practice (presentation and viva preparation)

Prepare spoken answers of three to five sentences each; brief guidance follows.

1. Describe the mill's complete process flow, including laboratory control points, without looking at your slides.
2. Take one knitting machine from your report: give its diameter, gauge and fabric, and explain how loop length affects GSM.
3. Explain right-first-time in dyeing: how the mill pursued it, its reported rate if disclosed, and what reprocessing costs.
4. The examiner points to any laboratory result: what test is it, by what standard, what was the limit, and what happens on failure?
5. Show how you calculated one efficiency figure, step by step, and judge whether the result is good.
6. What does the mill's ETP do, and how do audits connect the ETP, the laboratory and the order book?

Answer key

1. Follow the material in the mill's real order (Figure 1 adapted), naming the test releasing material at each stage. Fluency proves the visit was yours; practise a 60-second summary.
2. Quote the machine as recorded, then explain: a longer loop puts more yarn in each course, so the fabric is heavier, looser and more extensible; a shorter loop gives a tighter, more stable fabric. Link it to a GSM you collected.
3. RFT means passing shade and fastness approval first time, pursued by accurate dosing, controlled liquor ratio and profile, dip approval and consistent pretreatment. A failed lot costs dyes, water, steam, time and delay. If not disclosed, say so.
4. Give test, instrument, standard and limit as one chain — e.g. unevenness against the mill's U% limit — and the consequence: retest, rejection or downgrading. Never upgrade a reported limit into a measured result.
5. Work it as written: theoretical output from speed and specification, actual output as counted or reported, efficiency as their ratio × 100. Judge it by stops and changeovers, not against 100%.
6. The ETP equalises and neutralises effluent, removes colour, chemicals and solids, and discharges only water within pH, BOD and COD limits. A failed audit can suspend orders.